

► REPRODUCTIVE—PHYSIOLOGY

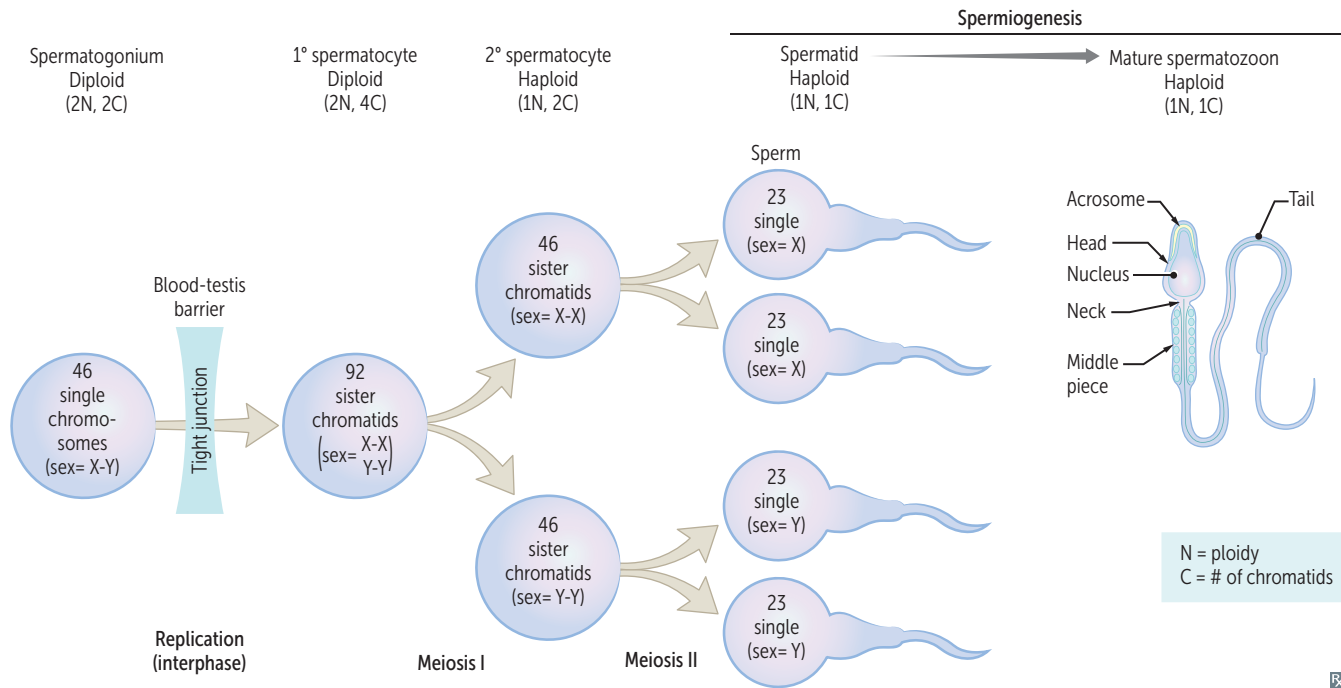
Spermatogenesis

Begins at puberty with spermatogonia. Full development takes 2 months. Occurs in seminiferous tubules. Produces spermatids that undergo spermiogenesis (loss of cytoplasmic contents, gain of acrosomal cap) to form mature spermatozoa.

“**G**onium” is **g**oing to be a sperm; “**z**oon” is “**z**ooming” to egg.

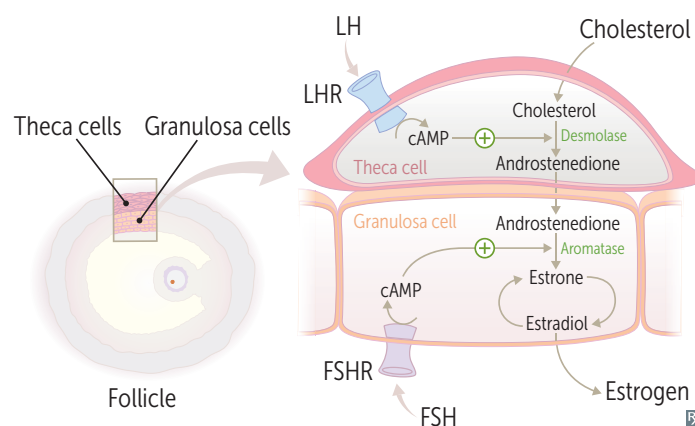
Tail mobility impaired in ciliary dyskinesia/Kartagener syndrome → infertility.

Tail mobility normal in cystic fibrosis (in CF, absent vas deferens → infertility).



Estrogen

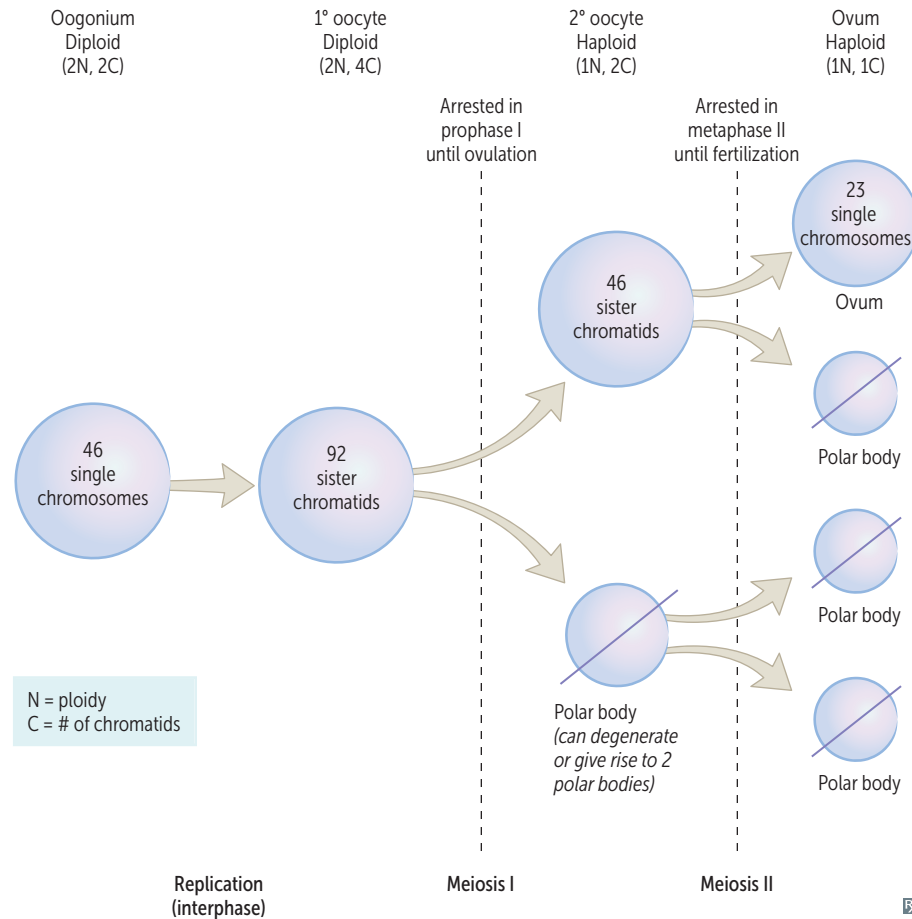
SOURCE	Ovary (estradiol), placenta (estriol), adipose tissue (estrone via aromatization).	Potency: estradiol > estrone > estriol. Estradiol is produced from 2 ovaries.
FUNCTION	Development of internal/external genitalia, breasts, female fat distribution. Growth of follicle, endometrial proliferation, ↑ myometrial excitability. Upregulation of estrogen, LH, and progesterone receptors; feedback inhibition of FSH and LH, then LH surge; stimulation of prolactin secretion, ↓ prolactin action on breasts. ↑ transport proteins, SHBG; ↑ HDL; ↓ LDL.	Pregnancy: 50-fold ↑ in estradiol and estrone 1000-fold ↑ in estriol (indicator of fetal well-being) Estrogen receptors expressed in cytoplasm; translocate to nucleus when bound by estrogen.

**Progesterone**

SOURCE	Corpus luteum, placenta, adrenal cortex, testes.	Fall in estrogen and progesterone after delivery disinhibits prolactin → lactation. ↑ progesterone is indicative of ovulation.
FUNCTION	During luteal phase, prepares uterus for implantation of fertilized egg: - Stimulation of endometrial glandular secretions and spiral artery development - Production of thick cervical mucus → inhibits sperm entry into uterus - Prevention of endometrial hyperplasia ↑ body temperature ↓ estrogen receptor expression ↓ gonadotropin (LH, FSH) secretion During pregnancy: - Maintenance of endometrial lining and pregnancy ↓ myometrial excitability → ↓ contraction frequency and intensity ↓ prolactin action on breasts	Progesterone is pro-gestation. Prolactin is pro-lactation.

Oogenesis

1° oocytes begin meiosis I during fetal life and complete meiosis I just prior to ovulation. Meiosis I is arrested in **prophase I** (one) for years until **ovulation** (1° oocytes). Meiosis II is arrested in **metaphase II** (two) until **fertilization** (2° oocytes). If fertilization does not occur within 1 day, the 2° oocyte degenerates.

**Ovulation**

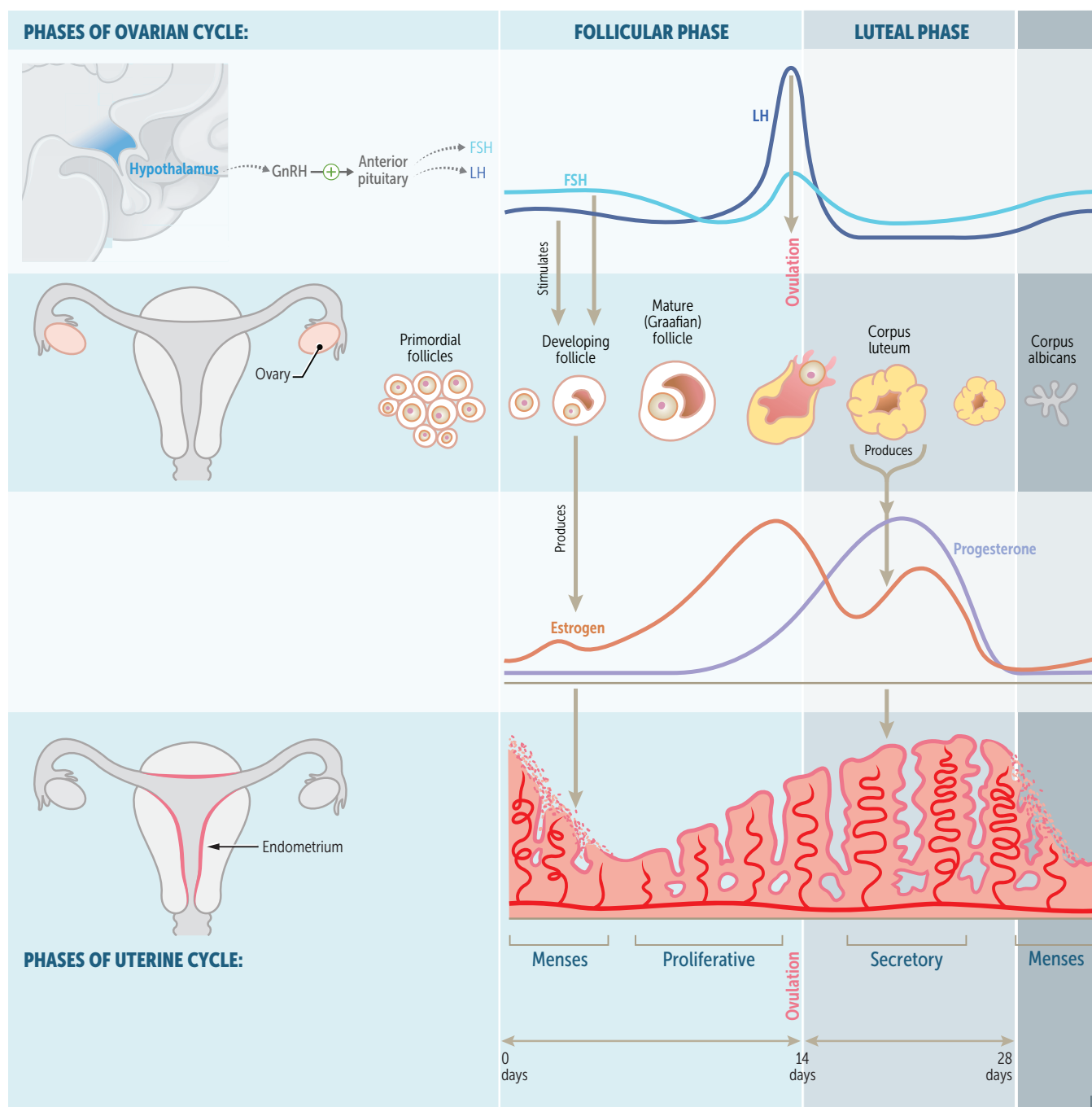
Follicular rupture and 2° oocyte release. Caused by sudden LH release (LH surge) at **midcycle**. Estrogen normally inhibits LH release, but high estrogen at midcycle transiently stimulates LH release → LH surge → ovulation.

Mittelschmerz (“**middle** hurts”)—pain with ovulation. Associated with peritoneal irritation from normal bleeding upon follicular rupture. Typically unilateral and mild, but can mimic acute appendicitis.

Menstrual cycle

Regular cyclic changes periodically preparing the female reproductive system for fertilization and pregnancy. Occurs in phases based on events taking place in ovaries and uterus.

	1 ST DAY OF MENSES TO OVULATION	OVULATION TO 1 ST DAY OF NEXT MENSES
Ovarian cycle	Follicular phase —follicular development; late stages are stimulated by FSH ; can fluctuate in length.	Luteal phase —corpus luteum formation from follicular remnants; stimulated by LH ; lasts a fixed 14 days.
Uterine cycle	Proliferative phase —endometrial development; stimulated by estrogen. Straight, narrow endometrial glands.	Secretory phase —endometrial preparation for implantation; stimulated by progesterone. Tortuous, dilated endometrial glands.



Abnormal uterine bleeding

Deviation from normal menstruation volume, duration, frequency, regularity, or intermenstrual bleeding.

Causes (**PALM-COEIN**):

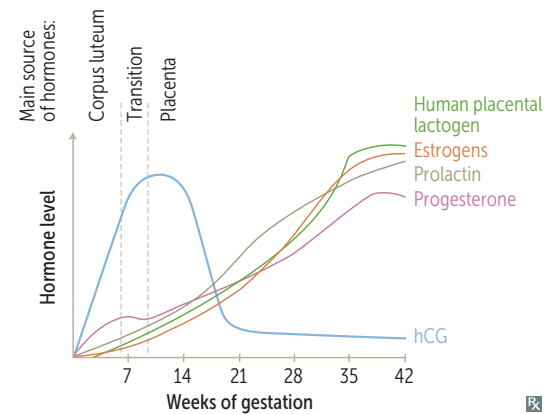
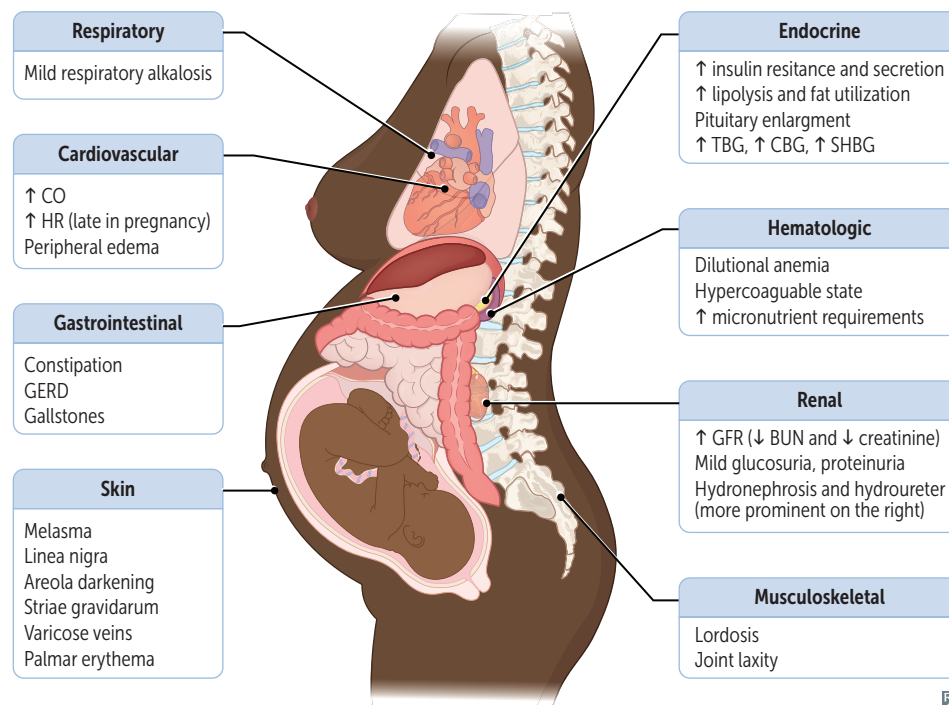
- Structural: **P**olyp, **A**denomyosis, **L**eiomyoma, **M**alignancy/hyperplasia
- Nonstructural: **C**oagulopathy, **O**vulatory, **E**ndometrial, **I**atrogenic, **N**ot yet classified

Terms such as dysfunctional uterine bleeding, menorrhagia, metrorrhagia, polymenorrhea, and oligomenorrhea are no longer recommended.

Pregnancy

Fertilization (conception) most commonly occurs in upper end of fallopian tube (the ampulla). Occurs within 1 day of ovulation. Implantation in the uterine wall occurs 6 days after fertilization. Syncytiotrophoblasts secrete hCG, which is detectable in blood 1 week after fertilization and on home urine tests 2 weeks after fertilization. Embryonic/developmental age—time since fertilization. Used in embryology. Gestational age—time since first day of last menstrual period. Used clinically. Gravidity (“gravida”)—number of pregnancies. Parity (“para”)—number of pregnancies that resulted in live births.

Placental hormone secretion generally increases over the course of pregnancy, but hCG peaks at 8–10 weeks of gestation.

**Physiologic changes in pregnancy**

Human chorionic gonadotropin

SOURCE	Syncytiotrophoblast of placenta.
FUNCTION	Maintains corpus luteum (and thus progesterone) for first 8–10 weeks of gestation by acting like LH (otherwise no luteal cell stimulation → abortion). Luteal-placental shift is complete after 8–10 weeks; placenta synthesizes its own estriol and progesterone and corpus luteum degenerates. Used to detect pregnancy because it appears early in urine (see above). Has identical α subunit as LH, FSH, TSH. β subunit is unique (detected by pregnancy tests) but structurally similar to that of TSH (states of $\uparrow$ hCG can cause hyperthyroidism). hCG is $\uparrow$ in multifetal gestation, hydatidiform moles, choriocarcinomas, and Down syndrome; hCG is $\downarrow$ in ectopic/failing pregnancy, Edwards syndrome, and Patau syndrome.

Human placental lactogen

Also called human chorionic somatomammotropin.

SOURCE	Syncytiotrophoblast of placenta.
FUNCTION	Promotes insulin resistance to supply growing fetus with glucose and amino acids. Concurrently stimulates insulin secretion; inability to overcome insulin resistance → gestational diabetes.

Apgar score

	Score 2	Score 1	Score 0
A pppearance	 Pink	 Extremities blue	 Pale or blue
P ulse	≥ 100 bpm	< 100 bpm	No pulse
G rimace	Cries and pulls away	Grimaces or weak cry	No response to stimulation
A ctivity	 Active movement	 Arms, legs flexed	 No movement
R espiration	Strong cry	Slow, irregular	No breathing

Assessment of newborn vital signs following delivery via a 10-point scale evaluated at 1 minute and 5 minutes. **A**pgar score is based on **a**pppearance, **p**ulse, **g**rimace, **a**ctivity, and **r**espiration. Apgar scores < 7 may require further evaluation. If Apgar score remains low at later time points, there is $\uparrow$ risk the child will develop long-term neurologic damage.

Neonatal birth weight

	Low birth weight	High birth weight (macrosomia)
DEFINITION	Birth weight < 2500 g	Birth weight > 4000 g
RISK FACTORS	Prematurity, FGR	Fetal: constitutional/genetic Maternal: obesity, diabetes mellitus
COMPLICATIONS	$\uparrow$ mortality (SIDS), $\uparrow$ morbidity	$\uparrow$ risk of maternal or fetal trauma (eg, shoulder dystocia)

Lactation

After parturition and delivery of placenta, rapid ↓ in estrogen and progesterone disinhibits prolactin → initiation of lactation. Suckling is required to maintain milk production and ejection, since ↑ nerve stimulation → ↑ oxytocin and prolactin.

Prolactin—induces and maintains lactation and ↓ reproductive function.

Oxytocin—assists in milk letdown; also promotes uterine contractions.

Breast milk is the ideal nutrition for infants < 6 months old. Contains immunoglobulins (conferring passive immunity; mostly IgA), macrophages, lymphocytes. Breast milk reduces infant infections and is associated with ↓ risk for child to develop asthma, allergies, diabetes mellitus, and obesity. Exclusively breastfed infants should get vitamin D +/- iron supplementation.

Breastfeeding ↓ maternal risk of breast and ovarian cancer and facilitates mother-child bonding.

Menopause

Diagnosed by amenorrhea for 12 months.

↓ estrogen production due to age-linked decline in number of ovarian follicles. Average age at onset is 51 years (earlier in people who smoke tobacco).

Usually preceded by 4–5 years of abnormal menstrual cycles. Source of estrogen (estrone) after menopause becomes peripheral conversion of androgens, ↑ androgens → hirsutism.

↑↑ FSH is specific for menopause (loss of negative feedback on FSH due to ↓ estrogen).

Hormonal changes: ↓ estrogen, ↑↑ FSH, ↑ LH (no surge), ↑ GnRH.

Causes **HAVOCS**: **H**ot flashes (most common), **A**trophy of the **V**agina, **O**steoporosis (↑ osteoclast activity), **C**oronary artery disease, **S**leep disturbances.

Hormone replacement therapy (HRT) is used to treat menopausal symptoms, but can be associated with ↑ risk of breast and endometrial cancer, CVD, stroke, and thromboembolism.

Menopause before age 40 suggests 1° ovarian insufficiency (premature ovarian failure); may occur in females who have received chemotherapy and/or radiation therapy.

Androgens

Testosterone, dihydrotestosterone (DHT), dehydroepiandrosterone (DHEA), androstenedione (adrenal).

SOURCE

DHT and testosterone (testis), DHEA and androstenedione (adrenal).

Potency: DHT > testosterone > androstenedione > DHEA.

FUNCTION

Testosterone:

- Differentiation of epididymis, vas deferens, seminal vesicles (internal genitalia, except prostate)
- Growth spurt: penis, seminal vesicles, sperm, muscle, RBCs
- Deepening of voice
- Closing of epiphyseal plates (via estrogen converted from testosterone)
- Libido

DHT:

- Early—differentiation of penis, scrotum, prostate
- Late—prostate growth, androgenetic alopecia, sebaceous gland activity

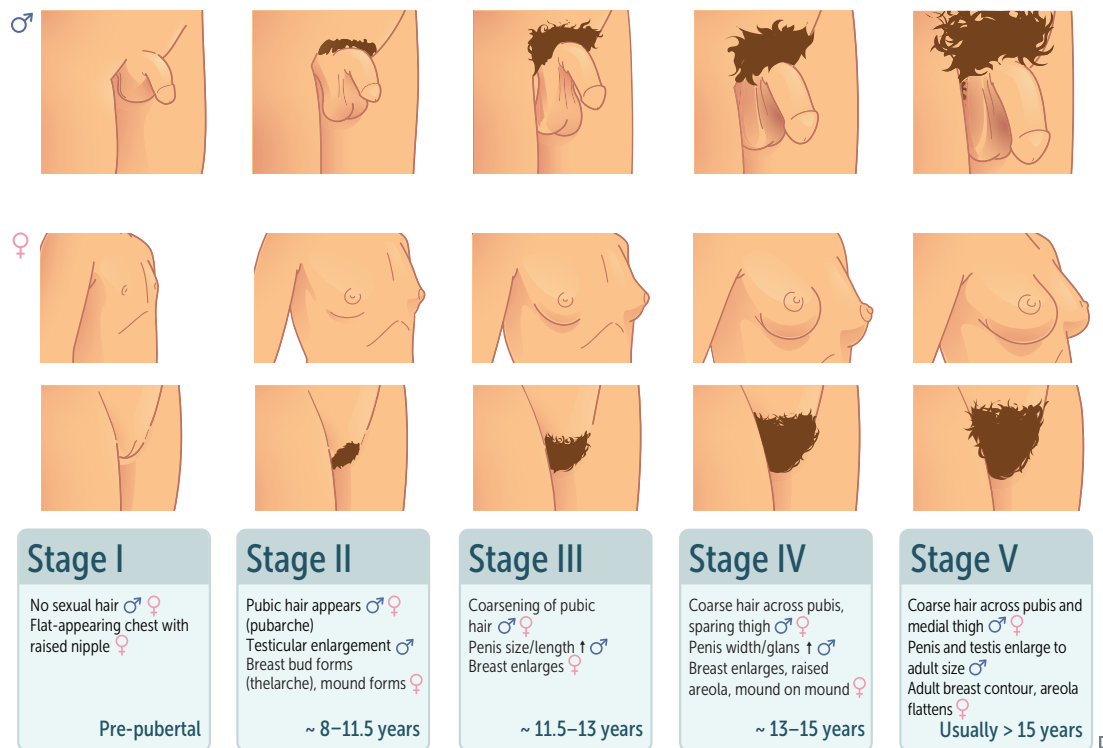
Testosterone is converted to DHT by 5α-reductase, which is inhibited by finasteride.

In the male, androgens are converted to estrogens by aromatase (primarily in adipose tissue and testes).

Anabolic-androgenic steroid use—↑ fat-free mass, muscle strength, performance. Suspect in males who present with changes in behavior (eg, aggression), acne, gynecomastia, erythrocytosis (↑ risk of thromboembolism), small testes (exogenous testosterone → hypothalamic-pituitary-gonadal axis inhibition → ↓ intratesticular testosterone → ↓ testicular size, ↓ sperm count, azoospermia). Females may present with virilization (eg, hirsutism, acne, breast atrophy, androgenetic alopecia).

Sexual maturity rating (Tanner stage) of pubertal development

Sexual maturity rating is assigned independently to genitalia, pubic hair, and breast (eg, a person can have stage 2 genitalia, stage 3 pubic hair). Earliest detectable secondary sexual characteristic is breast bud development in females, testicular enlargement in males.



Precocious puberty

Appearance of 2° sexual characteristics (eg, pubarche, thelarche) before age 8 years in females and 9 years in males. ↑ sex hormone exposure or production → ↑ linear growth, somatic and skeletal maturation (eg, premature closure of epiphyseal plates → short stature). Types include:

- Central precocious puberty (↑ GnRH secretion): idiopathic (most common; early activation of hypothalamic-pituitary gonadal axis), CNS tumors.
- Peripheral precocious puberty (GnRH-independent; ↑ sex hormone production or exposure to exogenous sex steroids): congenital adrenal hyperplasia, estrogen-secreting ovarian tumor (eg, granulosa cell tumor), Leydig cell tumor, McCune-Albright syndrome.

Delayed puberty

Absence of 2° sexual characteristics by age 13 years in females and 14 years in males. Causes:

- Hypergonadotropic (1°) hypogonadism: Klinefelter syndrome, Turner syndrome, gonadal injury (eg, chemotherapy, radiotherapy, infection).
- Hypogonadotropic (2°) hypogonadism: constitutional delay of growth and puberty (“late blooming”), Kallmann syndrome, CNS lesions.